

Operation Handwriting, Teacher's Guide

Read this once before Day 1. It takes 20 minutes. Everything you need each day is then in the daily plan (file 01) and the materials folder.

1. What we are fixing and why it went wrong

Bad handwriting in a school is almost never a talent problem. It comes from four habits that were never corrected: a wrong pencil grip, letters formed with the wrong stroke order (so they come out different every time), no sense of size (tall letters, small letters and tail letters all the same height), and rushing. Each of these is fixable in days with daily correction. The program works because for 10 days the same adult corrects the same habit in the same student until it is gone.

Your job for the two weeks is not to get through the worksheets. It is to watch hands and correct habits. A student who traces a full page with a fist grip has gained nothing.

2. The four non-negotiables you enforce every single day

Posture. Feet flat on the floor, back touching the chair, both forearms on the desk, the page tilted about 20 degrees (top of the page to the right for right-handers, to the left for left-handers), and the helper hand holding the page still. Say "feet, back, arms, paper, helper hand" and wait until every student has done it. This takes 30 seconds and it changes the writing.

Grip. Tripod grip: the pencil rests on the middle finger, held between the thumb pad and the tip of the index finger, about 2 to 3 cm from the point, the pencil resting in the web between thumb and index. Fingers slightly bent, not white-knuckled. A student who wraps the thumb over the index, holds in a fist, or uses four fingers gets a triangular grip on the pencil from Day 1 and you check it every day. Do not accept "it feels weird". It feels weird for three days and then it does not.

Pressure. The line should be a clear, even grey. If the page has grooves on the back, the student is pressing too hard: tell them to write as if the pencil could break. If the line is faint and wobbly, they are too tense in the fingers and holding too far from the tip. Put a sheet of paper behind the page and check the back.

Stroke order. Every letter has one way to be made. The verbal pathway (section 5) is said out loud by you and by the student while forming the letter. Students who "draw" letters (a circle then a stick for a, an o with a tail for g, a b starting from the bottom) will always be slow and inconsistent. In the two weeks we replace drawing with writing.

3. How to run the hour

The block plan is in file 01. Three things make or break it.

Model slowly and out loud. When you write on the board, write large, write slowly, and say the pathway as you do it. Then do it again. Then a third time with students saying it with you. Students then write it in the air with a straight arm (big movement builds the motor memory), then on the desk with a finger, then on paper. Do not skip the air-writing, the older ones will complain, do it anyway.

Circulate with a purpose. During guided practice you walk the room in a fixed loop and you look at hands first, page second. Order of correction: posture, grip, then formation. Fix one thing per student per pass. Touch the page and put a small dot where the next letter should start if the student keeps starting in the wrong place.

Praise the specific, not the general. "Good job" teaches nothing. "Your a starts at the top now, that is why it is closing properly" teaches everything. Every day, every student circles their own best letter or word and you write

one star (what improved) and one wish (the one thing to fix tomorrow) on the page. Keep the wish the same until it is fixed.

4. Marking policy

You do not correct every letter on a page. It crushes the student and tells them nothing. On each sheet you mark at most one star and one wish. The student's own circle of their best letter is the main self-assessment.

Rewriting: a page that is clearly rushed (compared to what the student produced the previous day) is handed back with "this is not your best, do the last two lines again". Not a punishment, a standard.

Never write on a student's work in red for this program. Use pencil or green.

5. Verbal pathways for print (Cycle 1, and print repair in Cycle 2)

Say these out loud while modelling. The student repeats them while writing. Every lowercase letter starts at the top except d and e. Letter families are taught together because the motor movement is the same.

Straight line family, taught Day 2. l: start at the top, straight down to the line. i: start at the middle, straight down, lift, dot on top. t: start at the top, straight down, lift, cross at the middle.

Round family, taught Day 3. All start at the two o'clock position and go anticlockwise. c: start just below the top of the small letters, curve up, around and down, stop before you close. o: same as c, keep going all the way round and close. a: make a c, close it, then straight down the right side to the line. d: make a c, then go up tall to the top, then straight down to the line. g: make a c, close it, down the right side under the line, hook to the left. q: make a c, close it, down under the line, small flick to the right.

Bump family, taught Day 4. Start at the top, down, then trace back up and over. r: down, back up, small bump over to the right and stop. n: down, back up, over the bump and down to the line. m: down, back up, bump, bump, down to the line. h: tall line down, back up halfway, bump over and down. b: tall line down, back up halfway, bump around and close on the line. p: down under the line, back up, bump around and close at the line.

Tricky family, taught Day 5. e: start in the middle, short line to the right, curve up and round like a c. s: start at the top, curve left like a small c, then swing back the other way to the line. f: start at the top, curve left, straight down to the line, lift, cross at the middle. k: tall line down, lift, small slanted line in, small slanted line out. u: down, curve at the bottom, up, then straight down. v: slant down, slant up. w: slant down, up, down, up. x: slant down right, lift, slant down left. y: slant down right to the line, lift, slant down left under the line. z: across, slant down, across. j: start at the middle, straight down under the line, hook left, lift, dot.

Sizes. Tall letters (b, d, f, h, k, l, t) touch the top line. Small letters (a, c, e, i, m, n, o, r, s, u, v, w, x, z) live between the middle line and the baseline. Tail letters (g, j, p, q, y) sit on the baseline and their tail goes down. On the three-line paper the students say "tall, small, or tail?" before writing a letter.

Capitals all start at the top and all are tall. Teach them by stroke type: straight lines (L, I, T, H, E, F), round (C, O, Q, G), lines plus curves (B, D, P, R, J, U), slants (A, V, W, X, Y, Z, K, M, N), and S.

6. Verbal pathways for cursive (Cycle 2 from Day 3, Cycle 3 repair)

Two ideas make cursive work and both are taught on Day 3 before any letter. The entry stroke: every lowercase letter begins with a short upward stroke from the baseline. The exit stroke: every letter ends with a short upward flick to the right that becomes the entry stroke of the next letter. Students who learn letters without entry and exit strokes cannot join them later.

Undercurve family, Day 3 (entry is a curve up from the line). i: curve up to the middle, straight down, exit flick, dot. u: curve up, down, curve at the bottom, up, down, exit. w: curve up, down, curve, up, down, curve, up, small loop, exit. t: curve up tall, straight down, exit, cross.

Loop family, Day 4 (the upstroke becomes a loop at the top). e: curve up, loop over, down and round, exit. l: curve up tall, loop, straight down, exit. h: curve up tall, loop, down, back up halfway, bump, down, exit. k: curve up tall, loop, down, back up, small loop, kick out, exit. b: curve up tall, loop, down, curve at the bottom, up to the middle, small hook, exit from the middle. f: curve up tall, loop, straight down under the line, loop back, cross at the line, exit.

Round family, Day 5 (entry curve then an anticlockwise circle). a: curve up, round anticlockwise and close, down the right side, exit. d: curve up, round and close, up tall, down, exit. g: curve up, round and close, down under the line, loop, exit. q: curve up, round and close, down under the line, loop to the right, exit. c: curve up, round anticlockwise, stop open, exit. o: curve up, round, close at the top, small hook, exit from the top.

Overcurve family, Day 6 (entry curves over the top). n: curve over, down, back up, over, down, exit. m: curve over, down, up, over, down, up, over, down, exit. v: curve over, down, curve up, small hook, exit from the top. x: curve over, down, exit, lift, small slant crossing. y: curve over, down, up, down under the line, loop, exit. z: curve over, down with a small curve, loop under the line, exit.

Remaining letters, Day 7. r: curve up, small shoulder to the right, down, exit. s: curve up, small loop at the top, down and round to the left, close at the line, exit. p: curve up, down under the line, back up, bump round, exit. j: curve up, down under the line, loop, exit, dot.

Joins. There are two: joins from the baseline (most letters) and joins from the top (after b, o, v, w). Practise in this order: same-letter pairs (ll, ee, nn), baseline joins (in, at, up, an), top joins (ov, wi, bo, wa, ve), then the four words on the joins sheet. A join is correct when the pencil never lifts inside the word.

Cursive capitals are not joined to the next letter unless it is natural. Teach the ones on the sheet, do not spend Day 8 on Q and Z.

7. Common problems and the fix

What you see	Cause	Fix
Letters float above the line or sink below	Student does not look at the baseline	Highlight the baseline in yellow on their sheet for two days. Say "sit it on the line".
All letters the same height	No concept of tall, small, tail	Size sort sheet, and the student says "tall, small or tail?" before each letter for two days.
Reversed b and d, p and q	Motor plan not automatic	Teach b as "line then bump" and d as "c then line". Never teach them on the same day.
Letters drawn in pieces, inconsistent	Wrong stroke order	Air-writing with the pathway, then trace three times saying it.
Words run together	No spacing habit	Finger space between words for Cycle 1, "one letter width" for older. Spacing sheet.
Page has grooves, hand aches	Too much pressure	Write on a soft pad of paper, "the pencil could break". Softer pencil (2B) for one week.
Faint, shaky line	Tension, holding too far up	Move fingers closer to the point, grip check, shake hands out between lines.
Slant changes within a word	Paper flat and straight on the desk	Tilt the paper, tape a strip at the correct angle for the week.
Fast and messy	Rushing, no standard	Set the standard once (rewrite the last two lines), timed work only from Day 7.
Cursive letters not joined	Missing exit strokes	Back to entry and exit strokes on Day 3 material for one session.

Left-hander smudging or hooking the wrist	Wrong paper position	See section 8.
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8. Left-handed students

Identify them on Day 1 and note them on the tracking sheet. They tilt the paper the opposite way (top to the left), so the hand stays below the writing line and does not cover what they wrote. They hold the pencil a little higher, 3 to 4 cm from the point, so they can see the letters. They sit on the left side of a shared desk so elbows do not collide. They never copy from a model placed on their left, put the model on the right or above. Do not correct a left-hander's slant to match right-handers, a slight backward slant is normal. Left-handers doing cursive may need a pencil that is not too soft, otherwise smudging discourages them.

9. Differentiation

The weakest five in each group, identified on Day 1 by the pre-test, work at a separate table during guided practice with the assistant or with you while the group is on the worksheet. They use the Cycle 1 wide lined paper regardless of their age, they do fewer letters per day, and the goal for them is grip, size and baseline only. Do not put them on cursive until print is legible.

Students with a diagnosed or suspected motor difficulty (dyspraxia, hypotonia, a student who cannot hold a tripod grip after a week of trying) are not failing the program. Note them on the tracking sheet, keep them on the wide paper, give them the pencil grip and the slant board, and send the name to the principal on Day 10 so the follow-up can involve an occupational therapy assessment if the parents agree.

Students who are already strong get the same first 15 minutes, then during guided practice they work on speed with legibility (Cycle 3 speed sheet), on the presentation sheet, or on copying a longer passage in their best script. They can also be paired with a weaker student for the applied writing block, but only to model, not to correct.

10. Scoring the pre-test and post-test

Same passage on Day 1 and Day 10, same time limit (Cycle 1: 10 minutes for the copy, Cycle 2 and 3: 8 minutes for the copy plus 5 minutes free writing). Do not help during the test. Score with the rubric in the materials file, six criteria, 1 to 4 each, 24 maximum. Score the pre-test the same day before you have seen the student write anything else, so the score is honest. Enter both scores on the class tracking sheet and hand it to the principal on Day 10 with the names of every student under 12 on the post-test.

Show each student their two samples side by side on Day 10. For most students this is the most convincing moment of the two weeks.

11. What to send home

Nothing is compulsory at home. On Day 2 send home the parent note from the materials file (the school announcement). If a parent asks what to do, the answer is: 5 minutes of copying a sentence in the student's best writing, with the paper tilted and the correct grip, and no correcting by the parent beyond posture and grip.

12. Your checklist for Day 1

Before the students arrive: pre-test sheets counted, lined paper stacked, progress charts and folders ready, warm-up card and posture poster on the wall, pencil grips for the students you already suspect, class tracking sheet with names filled in.

During: posture and grip routine taught and practised, pre-test done in silence, pattern strokes or self-check sheet, first star and wish given, progress chart box 1 coloured.

After: pre-tests scored and entered, five weakest names marked, left-handers marked.

